MAT 017B Midterm 1

August 16, 2018

Name: Key

SID:

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C,\ n \neq 1 \qquad \int \frac{1}{x}\,dx = \ln|x| + C \qquad \int e^x\,dx = e^x + C$$

$$\int \sin x\,dx = -\cos x + C \qquad \int \cos x\,dx = \sin x + C \qquad \int \sec^2 x\,dx = \tan x + C$$

$$\int \frac{1}{x^2+a^2}\,dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C \qquad \int_a^b f(x)\,dx = F(b) - F(a)$$

$$\int_a^b f(x)\,dx \approx \sum_{k=1}^{n} f\left(x_k\right)\Delta x, \quad \Delta x = \frac{b-a}{n}, \quad x_k = a + k\Delta x$$

$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x) \qquad \frac{d}{dx}\int_{h(x)}^{g(x)} f(t)\,dt = f[g(x)]g'(x) - f[h(x)]h'(x)$$

$$\int f'(g(x))g'(x)\,dx = f(g(x)) + C \qquad \int u\,dv = uv - \int v\,du$$

1. (10 points) Evaluate the following derivatives.

a. (5 points) $\dfrac{d}{dx}\displaystyle\int_{x\sin x}^{x^3} 3t\cos 5t\,dt$

$g(x) = x^3$ $\qquad$ $h(x) = x\sin x$

$g'(x) = 3x^2$ $\qquad$ $h'(x) = x\cos x + \sin x$

$$\boxed{\left[3x^3\cos(5x^3)\right]3x^2 - \left[3x\sin x\cos(5x\sin x)\right](x\cos x + \sin x)}$$

b. (5 points) $\dfrac{d}{dx}\displaystyle\int_{3}^{5x^2} \sqrt{5+e^{3t}}\,dt$

$$\boxed{10x\sqrt{5+e^{15x^2}}}$$

2. (40 points) Evaluate the following integrals.

a. (10 points) $\int \left(10x^9 + 21x^6\right) e^{x^{10}+3x^7}\, dx$

$u = x^{10} + 3x^7$

$du = x^9 + 21x^6\, dx$

$\int e^u\, du = e^u + C$

$= \boxed{e^{x^{10}+3x^7} + C}$

b. (10 points) $\int (\cos x) \ln(\sin x + 4)\, dx$

$w = \sin x + 4$

$dw = \cos x\, dx$

$\int \ln w\, dw$

$u = \ln w \qquad dv = dw$

$du = \frac{1}{w} dw \qquad v = w$

$= w \ln w - \int dw$

$= w \ln w - w + C$

$= \boxed{(\sin x + 4) \ln(\sin x + 4) - (\sin x + 4) + C}$

c. (10 points) $\int xe^x \, dx$

$u = x \qquad dv = e^x dx$

$du = dx \qquad v = e^x$

$= xe^x - \int e^x dx$

$= \boxed{xe^x - e^x + C}$

d. (10 points) $\int 4x \ln x \, dx$

$u = \ln x \qquad dv = 4x \, dx$

$du = \frac{1}{x} dx \qquad v = 2x^2$

$= 2x^2 \ln x - \int 2x \, dx$

$= \boxed{2x^2 \ln x - x^2 + C}$

3. (15 points) Approximate the following integral using the right endpoint rule with 4 intervals.

$$\int_{-2}^{6} x^2 + 2x + 4\, dx$$

$\Delta x = \dfrac{6-(-2)}{4} = 2$

$x_1 = -2 + 1 \cdot 2 = 0$

$x_2 = -2 + 2 \cdot 2 = 2$

$x_3 = -2 + 3 \cdot 2 = 4$

$x_4 = -2 + 4 \cdot 2 = 6$

$A_1 = 2(0^2 + 2 \cdot 0 + 4) = 2(4) = 8$

$A_2 = 2(2^2 + 2 \cdot 2 + 4) = 2(4+4+4) = 2(12) = 24$

$A_3 = 2(4^2 + 2 \cdot 4 + 4) = 2(16 + 8 + 4) = 2(28) = 56$

$A_4 = 2(6^2 + 2 \cdot 6 + 4) = 2(36 + 12 + 4) = 2(52) = 104$

$$\begin{array}{r} {}^{2}\\ 104 \\ 56 \\ 24 \\ +\quad 8 \\ \hline \boxed{192} \end{array}$$

4. (20 points) Evaluate the following integral.

$$\int \frac{3x^2+4x+8}{(x^2+2x+2)(x-1)}\,dx$$

$$\frac{3x^2+4x+8}{(x^2+2x+2)(x-1)} = \frac{Ax+B}{x^2+2x+2} + \frac{C}{x-1}$$

$$3x^2+4x+8 = (Ax+B)(x-1) + C(x^2+2x+2)$$

$$= Ax^2 - Ax + Bx - B + Cx^2 + 2Cx + 2C$$

$$A + C = 3 \quad \Rightarrow A = 3 - C = 3 - 3 = 0$$

$$-A + B + 2C = 4$$

$$-B + 2C = 8 \quad \Rightarrow B = 2C - 8 = 2(3) - 8 = -2$$

$$-(3-C) + 2C - 8 + 2C = 4$$

$$-3 + C + 2C - 8 + 2C = 4$$

$$5C = 15$$

$$C = 3$$

$$\int \frac{-2}{x^2+2x+2}\,dx + \int \frac{3}{x-1}\,dx$$

$$\int \frac{-2}{(x+1)^2+1}\,dx + \int \frac{3}{x-1}\,dx$$

$$= \boxed{-2\tan^{-1}(x+1) + 3\ln|x-1| + C}$$

5. (15 points) Evaluate the following integral.

$$\int_0^\infty x^2 e^{-x}\, dx$$

Hint: Use L'Hôpital's Rule $\left(\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to a}\frac{f'(x)}{g'(x)}\right)$ to evaluate the limit.

$$\lim_{a\to\infty}\int_0^a x^2e^{-x}\,dx = \lim_{a\to\infty}\left[-x^2e^{-x}-2xe^{-x}-2e^{-x}\right]_0^a$$

$$= \lim_{a\to\infty}\left[\cancel{-a^2e^{-a}}^{\,0}-\cancel{2ae^{-a}}^{\,0}-\cancel{2e^{-a}}^{\,0}+0^2e^{-0}+2\cdot 0e^{-0}+2e^{-0}\right]$$

$$= \boxed{2}$$

	u	dv
$+$	x^2	e^{-x}
$-$	$2x$	$-e^{-x}$
$+$	2	e^{-x}
		$-e^{-x}$

$$\lim_{a\to\infty} -a^2e^{-a} = \lim_{a\to\infty}\frac{-a^2}{e^a} = \lim_{a\to\infty}\frac{-2a}{e^a} = \lim_{a\to\infty}\frac{-2}{e^a} = \frac{-2}{\infty} = 0$$

$$\lim_{a\to\infty} -2ae^{-a} = \lim_{a\to\infty}\frac{-2a}{e^a} = \frac{-2a}{\infty} = 0$$

$$\lim_{a\to\infty} -2e^{-a} = 0$$